



HAMZA MASOOD

Third Year Undergraduate
Department of Electrical Engineering
Indian Institute of Technology Kanpur

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EDUCATION

Year	Qualification	School/Institute	CPI/%
2021 - Present	BTech	IIT Kanpur	9.8 /10.0
2021	Class XII (ISC)	Mercy Memorial School, Kanpur	99.8%
2019	Class X (ICSE)	Mercy Memorial School, Kanpur	98.4%

PROJECTS

Mathematical Trading Strategies

Stamatics, IIT Kanpur

May'23 - Jul'23

- Implemented technical indicators like **Keltner Channel**, **Bollinger Bands** and **MACD** for trading decisions
- Implemented **trading strategies** using these indicators and optimised the indicator parameters for NSE index after evaluating strategy performance and generated buy/sell signal for NASDAQ
- Automated detection of **chart patterns** such as ascending triangle, flags, head and shoulder and rounding top
- Designed** and implemented an effective trading strategy by combining technical indicators and patterns, specifically **Keltner Channel Bands** and **Hammer candlestick patterns**
- Applied statistical concepts and metrics like correlation coefficient, **Sharpe ratio**, and **Sortino ratio** for back-testing

Introduction to Game Development

SnT, IIT Kanpur

May'22 - Jul'22

- Developed John Lemon's Haunted Jaunt, a game as part of the initial evaluation task, utilizing **Unity** game engine and writing **C# scripts**, and added additional features ([Link](#))
- Led a team of six members to design and develop an innovative **tower defense game** in Unity, contributing as a programmer
- Implemented various **game mechanics** including enemy movement, enemy wave spawner, shooting and targetting for weapons
- The game has **10 levels** and was published on [itch.io](#) under the name Tower Defence 101 ([Link](#))

Image Processing & Computer Vision

Stamatics, IIT Kanpur

May'23 - Jul'23

- Implemented image processing techniques such as image flipping and color channel manipulation to create a **jigsaw-solver**
- Created a **dot-matrix display** that displays any 2 digit number
- Applied image enhancement techniques like **Histogram equalization**, **blurring** and **filtering** to solve real world problems
- Developed a **convolutional neural network** model to classify document images into **16** different categories using **transfer learning** on the InceptionResNetV2 architecture
- Achieved a test accuracy of **95%**, with a validation accuracy of 70% by leveraging pre-trained weights and fine-tuning the model

WebDev & Unix Command Line

ACA, IIT Kanpur

Jun'22 - Aug'22

- Built a multi-page static website using the fundamentals of front-end tools such as **HTML**, **CSS**, and **Javascript**
- Acquired comprehensive understanding of the **Linux kernel**, Command Line, Version Control and Shell Scripting
- Completed the given assignments of the respective project weekly on **Ubuntu** OS using **Linux** Command Lines

RELEVANT COURSES

Probability & Statistics[†]

Linear Algebra & ODE

Partial Differential Equations[†]

Data Structure & Algorithm

Introduction to Electronics

Control System Analysis

Fundamentals of Computing

Microelectronics[†]

Introduction to Logic[†]

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 685**, in **Joint Entrance Exam Advanced 2021** among **150,000** shortlisted candidates
- Secured **All India Rank 524**, in **Joint Entrance Exam Mains 2021** among **1 million** candidates across the country
- Recipient of the **Quadeye Excellence Scholarship 2023** awarded to the **top 20** students of the batch
- Awarded **Dr. Prateek Mishra Memorial Scholarship** for having the **highest CPI** in EE Department
- Recipient of the **Inspire Scholarship** granted by the Govt. of India to the **top 1%** students in Class XII Board Examinations
- Received **Academic Excellence Award** in **2022** and **2023**, awarded by IIT Kanpur to the **top 10%** meritorious students
- Achieved a perfect **10** Semester Performance Index in the 2nd and 4th semesters, with **outstanding grade (A*)** in **6** courses

COMPETITIVE PROGRAMMING

- Competent in competitive programming with maximum ratings **1622** (Expert) on **Codeforces** (user handle - **struggler_op**) and **1981** on **Codechef** (user handle - **hamzam21**)
- Ranked **111**, in **ICPC Kanpur-Mathura 2022 Qualifier Round** among **2306** teams, and secured **2nd** position among teams from IIT Kanpur, Team Name - **Variables**

TECHNICAL SKILLS

- Languages:** C/C++ | Python | Java | C# | JavaScript | SQL
- Exposure:** Numpy | Pandas | Matplotlib | OpenCV | React
- Tools/Utilities:** Git | GitHub | Unity | $\LaTeX$

POSITIONS OF RESPONSIBILITY

Student Guide, Counselling Service Sep'22 – Present

- Responsible for ensuring a smooth transition of a group of **6** freshmen into college life and assisted the core team members in conduction of **orientation** for over **1000** UG students

Academic Mentor, Counselling Service Sep'22 – Jul'23

- Provided one-to-one mentorship and conducted academic classes for over **500** students in courses on **Mathematics**

Senior Executive, Techkriti'23 Aug'22 – Jan'23

- Designed **logical questions** and **puzzles** for both phases of Techkriti Open School Championship'23, while effectively coordinating the activities of over **50** Junior Executives

EXTRA CURRICULAR ACTIVITIES

- Coordinator, Card & Board Games Club IITK:** Organized and managed game nights and online events
- Secretary, Anime Society IITK:** Hosted and designed Anime Quizzes and created content for social media handles
- Secured 10th place among 200 teams in Mathemania 2022, a PAN IIT mathematics competition organized by Stamatics IITK

[†] : Outstanding Performance